CROSSSAFE · USER REQUIREMENTS

CrossSafe user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First user requirements from the users and stakeholders in CRS-PRB-001 (kit 1.9.1 upgrade); proposed, not yet validated with users or a road authority

CrossSafe user requirements

What users need, written from their side. Product requirements that meet these needs are in CRS-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix CRS-RTM-001 (`03b-traceability.md`). They are drawn from the users and stakeholders in CRS-PRB-001 and are proposals until they are checked with a partner (co-design, TRL 4). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Drivers notice the warning and yield	Drivers see the flashing indications early and in all light, so that they slow and yield to a waiting pedestrian	High	Pedestrian; Driver (PRB Table 1)
UR-002	Performance	Warning when someone waits, on both sides	The warning starts within half a second of a trigger on either side and only when someone is waiting or presses the button	High	Pedestrian; School or market community
UR-003	Usability	Anyone can operate it and see it is on	Children, older people and people with disabilities can reach and press the button, and know from sound, touch and a light that the warning is on	High	Pedestrian, including children, older people and people with disabilities
UR-004	Performance	The warning lasts long enough, and no longer	The warning runs long enough to cross at a slow walking pace and ends by itself, so that drivers do not learn to ignore it	High	Driver; Pedestrian
UR-005	Environment and durability	Works without mains and through poor-sun weeks	The crossing keeps working off the grid, in the worst sun month and through several dull days	High	School or market community
UR-006	Regulatory	Meets local rules and survives the weather	The device can be accepted by the road authority: right indication size and pattern, and it survives wind, heat, rain and dust for years	High	Road authority or municipal engineer
UR-007	Safety	A fault never leaves a misleading warning	Faults are shown and logged, and the device never flashes continuously or goes dark without notice	High	Road authority or municipal engineer
UR-008	Deployment and assembly	Fits an existing pole and installs quickly	A two-person crew fits it to a common pole in about two hours, without a bucket truck where possible	Medium	Road authority or municipal engineer; Local technician
UR-009	Cost	Costs little to buy	The parts for one assembly cost no more than the value-engineering target, so a school or market group can fund a crossing	High	School or market community; Road authority
UR-010	Maintenance	Built and repaired with ordinary tools and parts	A local workshop can make the made parts, and a technician can service and replace parts with hand tools and a multimeter	Medium	Local technician (PRB Table 1)
UR-011	Regulatory	No images or audio of people are kept	The device records and sends only counts, faults and battery state	Medium	Pedestrian; Road authority
UR-012	Environment and durability	Resists theft and vandalism	Batteries and electronics are out of casual reach and the fixings need tools	Medium	School or market community; Road authority

ID	Category	Title	Need	Priority	Source
UR-013	Safety	The lights are not harmful to look at	People standing beside the device or driving past are not dazzled or put at risk by the flashing	Medium	Pedestrian; Driver
UR-014	Performance	The warning is seen early enough to stop	At road speeds of 30 to 60 km/h a driver sees the warning at a distance longer than the stopping distance	High	Driver

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.